

The maximal-operator-space conjecture for compact-open groups

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Status. Candidate substantial partial result, likely valid, subject to human review. The theorem proves the source conjecture for locally compact groups possessing a compact open subgroup. The general connected noncompact case remains open.

1 Source conjecture

The source is N. Spronk [1]. The paper is titled *Operator space structure on Feichtinger's Segal algebra*. Its arXiv identifier is math/0607299v2. On source p. 17, after the Banach projective tensor formula, it conjectures that the canonical operator space on $S_0(G)$ is maximal exactly when G admits an open abelian subgroup. The source statement and its then-known compact-abelian-open case are reproduced below.

where we obtain the last identification by realising $T \times \{e_H\}$ as a transversal for the left cosets of $A \times H$ in $G \times H$. $\square$

We note that if G and H are both compact, neither having an open abelian subgroup, then the above result fails by Corollary 2.5(ii) and [18]. We conjecture that our operator space structure on $S_0(G)$ is the maximal operator space structure exactly when G admits an open abelian subgroup. This would imply the result above. However, it is clear, only when G has a compact abelian open subgroup, that our operator space structure on $S_0(G)$ is the maximal one. Indeed if G is abelian, then for an arbitrary closed ideal $\mathcal{I}$ of $A(G)$, i.e. of $L^1(\hat{G})$, it is not clear that the subspace operator space structure is the maximal one, whence we have no means to deduce that $S_0(G) \cong \ell^1(G) \hat{\otimes} \mathcal{I} / \ker q_{\mathcal{I}}$ is a maximal operator space. For results on subspaces of maximal operator spaces, see [20], for example.

The source's operator space is intrinsically determined only up to complete isomorphism [1, Introduction]. Accordingly, "maximal" below means completely isomorphic to the maximal operator space on the underlying Banach space.

2 Proof intuition

A compact open subgroup K turns the local question into a compact Fourier- algebra question. Restriction is a complete surjection $S_0(G) \rightarrow S_0(K)$, so maximality of $S_0(G)$ descends to $S_0(K)$. Compactness identifies the latter with $A(K)$. If $A(K)$ is maximal, its operator projective tensor square is just its Banach projective tensor square; but the source's functorial tensor theorem identifies the operator square with $A(K \times K)$. Losert's characterization then says that K is virtually abelian. Since K is open, the finite-index abelian subgroup it contains is open in G .

For the converse, an open abelian subgroup A meets K in a compact abelian open subgroup of G , exactly the sufficient case already proved in the source. The compact-open assumption is therefore used in a symmetric, structural way rather than as a technical separability condition.

3 Formal result

We use two standard operator-space facts [3]: quotients of maximal operator spaces are maximal, up to complete isomorphism; and

$$\text{MAX}(E) \widehat{\otimes} \text{MAX}(F) \cong \text{MAX}(E \otimes_{\gamma} F),$$

where $\widehat{\otimes}$ and $\otimes_{\gamma}$ denote the operator and Banach projective tensor products.

Theorem 1. *Let G be a locally compact group possessing a compact open subgroup. Then the canonical operator space structure on $S_0(G)$ is maximal if and only if G possesses an open abelian subgroup.*

Proof. Let K be a compact open subgroup of G .

Assume first that the canonical operator space on $S_0(G)$ is maximal. Source Theorem 3.3 says that restriction

$$R_K : S_0(G) \longrightarrow S_0(K), \quad u \longmapsto u|_K,$$

is a complete surjection. In the source's terminology this means that the induced map from $S_0(G)/\ker R_K$ onto $S_0(K)$ is a complete isomorphism. Since operator quotients of maximal spaces are maximal, $S_0(K)$ is maximal up to complete isomorphism.

Because K is compact, source Corollary 2.5(ii) gives a complete isomorphism

$$S_0(K) \cong A(K).$$

Thus $A(K)$ is maximal. The maximal tensor identity and source Theorem 3.1 now give, through the natural elementary-tensor map,

$$\begin{aligned} A(K) \otimes_{\gamma} A(K) &\cong A(K) \widehat{\otimes} A(K) \\ &\cong S_0(K) \widehat{\otimes} S_0(K) \\ &\cong S_0(K \times K) \\ &\cong A(K \times K). \end{aligned}$$

Losert's tensor-product theorem [2] states that the natural map $A(K) \otimes_{\gamma} A(K) \rightarrow A(K \times K)$ is an isomorphism only if K has an abelian subgroup B of finite index (equivalently, an open abelian subgroup). Since B is open in K and K is open in G , B is open in G . This proves necessity.

Conversely, suppose that G has an open abelian subgroup A . Then $A \cap K$ is a subgroup that is abelian, compact (as a closed subgroup of K), and open in G (as the intersection of two open subgroups). The paragraph following the source conjecture establishes that the canonical operator space on $S_0(G)$ is maximal whenever G has a compact abelian open subgroup. Applying that result to $A \cap K$ proves sufficiency. $\square$

Corollary 2. *The source conjecture holds for every totally disconnected locally compact group.*

Proof. By van Dantzig's theorem [4], such a group has a compact open subgroup, so Theorem 1 applies. $\square$

The same argument also records a useful necessary condition without a compact-open assumption on G itself.

Corollary 3. *If the canonical operator space on $S_0(G)$ is maximal, then every compact subgroup of G has an open abelian subgroup.*

Proof. For any compact subgroup $K \leq G$, repeat the restriction-quotient and Losert argument in the necessity half of Theorem 1. $\square$

4 Upgrade attempts and remaining obstruction

Eight focused routes were checked before stopping at the partial theorem. First, the compact-open restriction route yielded Theorem 1. Second, quotienting by compact normal subgroups only reproduces the same class. Third, unrestricted closed-subgroup descent gives Corollary 3, but does not characterize groups such as the affine or Heisenberg group, whose compact subgroups are abelian. Fourth, the source averaging map passes maximality to quotients, but available quotients of these connected models remain noncompact.

Fifth, a Gleason–Yamabe reduction moves the problem toward connected Lie groups but still leaves solvable nonabelian base cases. Sixth, maximality does give the Banach projective formula $S_0(G) \otimes_\gamma S_0(G) \cong S_0(G \times G)$; outside the compact case there is no justified passage from this identity to Losert’s identity for $A(G)$. Seventh, a compactly supported Fourier-algebra ideal would transfer the obstruction if it were completely complemented in $S_0(G)$; an exact idempotent cutoff is available for compact open cosets, bringing the argument back to Theorem 1, but not for a general connected group. Eighth, classical atomic decompositions in the abelian case do not automatically identify the source’s quotient matrix norms with maximal matrix norms.

Thus the unresolved part is not a missing group-theoretic reduction: it is the operator-space behavior of non-idempotent local ideals in groups with noncompact identity component. No claim is made for that range.

5 Novelty audit and verification recommendation

A bounded search on 13 August 2026 covered the exact title and arXiv id, “Feichtinger Segal algebra maximal operator space”, “open abelian subgroup”, “compact open subgroup”, and later work on p -Feichtinger algebras and algebraic characterizations. It located restatements of the source and related functorial results, but no later proof of this compact-open subcase or of the full conjecture. Novelty confidence is provisional because the proof is a short synthesis of source theorems and Losert’s classical characterization.

Human review should focus on two compatibility points: that “complete surjection” in source Theorem 3.3 is the complete-isomorphic quotient notion needed for maximality descent, and that the natural maps in the tensor chain are the elementary-tensor map appearing in Losert’s theorem. The source’s definitions and Theorem 3.1 make both identifications explicit.

References

- [1] N. Spronk, *Operator space structure on Feichtinger’s Segal algebra*, J. Funct. Anal. 248 (2007), 152–174; arXiv:math/0607299v2.
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- [3] E. G. Effros and Z.-J. Ruan, *Operator Spaces*, London Mathematical Society Monographs, New Series 23, Oxford University Press, 2000.
- [4] D. van Dantzig, *Zur topologischen Algebra. III. Brouwersche und Cantorsche Gruppen*, Compos. Math. 3 (1936), 408–426.